

HTML Tables

Introduction

An **HTML Table** is used to organize and display data in rows and columns, just like a table in a spreadsheet or database.

Example:

Name	Age	City
John	25	New York
Alice	22	London

In HTML, tables are created using the `<table>` element along with several other table-related tags.

Basic Structure of an HTML Table

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>

  <tr>
    <td>John</td>
    <td>25</td>
    <td>New York</td>
  </tr>

  <tr>
    <td>Alice</td>
    <td>22</td>
    <td>London</td>
  </tr>
</table>
```

Output Structure:

Name	Age	City
John	25	New York
Alice	22	London

HTML Table Tags

1. <table>

The `<table>` tag is the container that holds the entire table.

Syntax:

```
<table>
  ...
</table>
```

Example:

```
<table>
</table>
```

2. <tr> (Table Row)

The `<tr>` tag defines a table row.

Each `<tr>` creates one horizontal row.

Example:

```
<table>
  <tr>
    <td>Apple</td>
    <td>Mango</td>
  </tr>
</table>
```

Output:

```
-----  
| Apple | Mango |  
-----
```

3. `<td>` (Table Data)

The `<td>` tag represents a normal data cell.

Example:

```
<table>  
  <tr>  
    <td>John</td>  
    <td>25</td>  
  </tr>  
</table>
```

Output:

```
-----  
| John | 25 |  
-----
```

4. `<th>` (Table Header)

The `<th>` tag defines a header cell.

By default, browsers display header text in **bold** and **center-aligned**.

Example:

```
<table border="1">  
  <tr>  
    <th>Name</th>  
    <th>Age</th>  
  </tr>  
  
  <tr>  
    <td>John</td>  
    <td>25</td>  
  </tr>  
</table>
```

Output:

Name	Age
John	25

Table Attributes

1. border

The `border` attribute adds a border around the table.

Syntax:

```
<table border="1">
```

Example:

```
<table border="2">
  <tr>
    <td>A</td>
    <td>B</td>
  </tr>
</table>
```

2. width

Defines the width of the table.

Example:

```
<table border="1" width="500">
```

or

```
<table border="1" width="80%">
```

3. height

Defines the height of the table.

Example:

```
<table border="1" height="200">
```

Cell Padding

Definition

Cell Padding is the **space between the content and the border of a cell**.

Think of it as the **inner spacing** of a cell.

Example:

Without Cell Padding

```
-----  
| ABC |  
-----
```

With Cell Padding

```
-----  
|   ABC   |  
-----
```

Syntax:

```
<table border="1" cellpadding="20">
```

Example:

```
<table border="1" cellpadding="20">  
  <tr>  
    <td>Apple</td>  
    <td>Mango</td>  
  </tr>
```

</table>

Cell Spacing

Definition

Cell Spacing is the **space between two adjacent cells**.

Think of it as the **gap between cells**.

Without Cell Spacing

```
-----  
|A|B|C|  
-----
```

With Cell Spacing

```
|A|  |B|  |C|
```

Syntax:

```
<table border="1" cellspacing="10">
```

Example:

```
<table border="1" cellspacing="10">  
  <tr>  
    <td>A</td>  
    <td>B</td>  
  </tr>  
</table>
```

Difference Between Cell Padding and Cell Spacing

Cell Padding	Cell Spacing
--------------	--------------

Space inside a cell	Space between cells
Increases inner spacing	Increases gap between cells
Uses <code>cellpadding</code>	Uses <code>cellspacing</code>

Visualization:

Cell Padding

```

+-----+
|       |
|  Content  |
|       |
+-----+

```

Cell Spacing

```

+---+   +---+
| A |   | B |
+---+   +---+

```

colspan

Definition

`colspan` is used to merge multiple columns into one cell.

Syntax:

```
<td colspan="2">
```

Example:

```

<table border="1">
  <tr>
    <th colspan="2">Student Details</th>
  </tr>

  <tr>
    <td>Name</td>
    <td>John</td>
  </tr>
</table>

```

Output:

Student Details		
Name	John	

rowspan

Definition

`rowspan` merges multiple rows into one cell.

Syntax:

```
<td rowspan="2">
```

Example:

```
<table border="1">
  <tr>
    <td rowspan="2">John</td>
    <td>Math</td>
  </tr>
  <tr>
    <td>Science</td>
  </tr>
</table>
```

Output:

John	Math
	Science

<caption>

The `<caption>` tag provides a title for the table.

It appears above the table by default.

Example:

```
<table border="1">

<caption>Student Information</caption>

<tr>
  <th>Name</th>
  <th>Age</th>
</tr>

<tr>
  <td>John</td>
  <td>25</td>
</tr>

</table>
```

<thead>

Groups the header section of the table.

Example:

```
<table border="1">

<thead>
<tr>
  <th>Name</th>
  <th>Age</th>
</tr>
</thead>

<tbody>
<tr>
  <td>John</td>
  <td>25</td>
</tr>
</tbody>

</table>
```

<tbody>

<tbody>

Contains the main body of the table.

Example:

```
<tbody>
<tr>
  <td>Alice</td>
  <td>22</td>
</tr>
</tbody>
```

<tfoot>

Defines the footer section of the table.

Example:

```
<table border="1">

<tfoot>
<tr>
  <td>Total</td>
  <td>2 Students</td>
</tr>
</tfoot>

</table>
```

Complete Example

```
<!DOCTYPE html>
<html>

<head>
  <title>HTML Tables</title>
</head>

<body>

<table border="1"
  cellpadding="10"
  cellspacing="5"
```

```

width="500">

<caption>Student Marks</caption>

<thead>

<tr>
    <th rowspan="2">Name</th>
    <th colspan="2">Marks</th>
</tr>

<tr>
    <th>Math</th>
    <th>Science</th>
</tr>

</thead>

<tbody>

<tr>
    <td>John</td>
    <td>90</td>
    <td>85</td>
</tr>

<tr>
    <td>Alice</td>
    <td>95</td>
    <td>92</td>
</tr>

</tbody>

<tfoot>

<tr>
    <td colspan="3">Total Students: 2</td>
</tr>

</tfoot>

</table>

</body>

</html>

```

Output

Student Marks

Student Marks

Name	Marks	
	Math	Science
John	90	85
Alice	95	92
Total Students: 2		

How the Table Appears

- **Student Marks** is displayed as the **caption** above the table.
- The **Name** header spans **2 rows** using `rowspan="2"`.
- The **Marks** header spans **2 columns** using `colspan="2"`.
- The second header row contains **Math** and **Science**.
- The last row merges all three columns using `colspan="3"` to display **Total Students: 2** making it more readable and easier to maintain.